Defense Mechanisms: Architectural, Runtime, and Coordination Layers

This section presents the defense mechanisms comprising CIF. We begin with the cognitive security operator posture (`sec:operator-posture`), then organize specific defenses into three categories: architectural (`sec:arch-defenses`), runtime (`sec:runtime-defenses`), and coordination (`sec:coord-defenses`). We analyze defense composition (`sec:defense-composition`) and cost-benefit tradeoffs (`sec:cost-benefit`).

Cognitive Security Operator Posture

Cognitive Security Operator Posture I

Before examining specific defense mechanisms, we introduce the conceptual framework that guides their deployment: the **cognitive security operator posture**. This is the proactive defensive stance required when securing systems whose attack surface spans beliefs, goals, and inter-agent coordination.

Definition (Cognitive Security Operator Posture)

The cognitive security operator posture is a defensive configuration characterized by:

1. **Assume Breach**: Operate under the assumption that some cognitive states may already be compromised
2. **Defense in Depth**: Layer multiple independent defense mechanisms
3. **Continuous Verification**: Continuously verify beliefs, goals, and trust relationships rather than trusting initial state
4. **Graceful Degradation**: Maintain functionality under attack by isolating compromised components
5. **Observable Internals**: Make cognitive state inspectable for monitoring and forensics

Definition and Principles I

This posture differs fundamentally from traditional perimeter security, which assumes trusted internals protected by boundary defenses. In cognitive systems, the “perimeter’ ’ is the agent’s reasoning process itself—attacks can originate from legitimate, authenticated channels and manifest as corrupted beliefs rather than malformed packets.

The Observer Effect Challenge I

A distinct challenge in cognitive security is the **observer effect**: monitoring changes behavior. When agents know their beliefs are being monitored, several phenomena emerge:

The Observer Effect Challenge I

- **Adversarial adaptation**: Attackers modify payloads to avoid detection patterns
- **Stealth pressure**: Attacks become more subtle, trading impact for evasion
- **Monitoring overhead**: Continuous observation consumes resources and adds latency

The Observer Effect Challenge I

The operator posture embraces this dynamic rather than fighting it. By making monitoring visible and consistent, we shift the adversarial game toward smaller, slower attacks that our drift detection can identify over time (`sec:runtime-defenses`).

Traditional operational security (OPSEC) focuses on protecting information from adversaries. CogSec (cognitive security) extends this to protecting reasoning processes:

Cognitive Hygiene Practices:

1. **Belief Provenance**: Track the source of every high-confidence belief; reject beliefs without verifiable provenance
2. **Goal Anchoring**: Periodically reaffirm goals against original principal instructions; detect goal drift
3. **Trust Calibration**: Regularly recalibrate trust scores based on outcome verification; never assume trust stability
4. **Context Boundaries**: Enforce hard boundaries on context window usage; prevent unbounded context accumulation
5. **Memory Sanitization**: Audit and sanitize persistent memory stores; remove dormant injection payloads

Cognitive Compartmentalization:

$$\sigma_i^{\text{isolated}} = \langle \mathcal{B}_i^{\text{task}}, \mathcal{G}_i^{\text{task}}, \mathcal{I}_i^{\text{task}}, \mathcal{H}_i^{\text{task}} \rangle \tag{1}$$

Each task receives isolated cognitive state, preventing cross-contamination. A compromised task cannot pollute beliefs used by other tasks.

When cognitive attacks are detected, the response differs from traditional incident response:

Incident Response for Cognitive Attacks I

Table 1: Cognitive incident response escalation.

Level	Trigger	Response
L1	Single tripwire alert	Log, continue with heightened monitoring
L2	Multiple alerts or drift detection	Isolate affected agent, replay recent history
L3	Coordinated attack indicators	Pause delegation, require human approval
L4	Byzantine threshold exceeded	Halt consensus operations, enter safe mode
L5	Principal goal corruption detected	Full system halt, require principal re-authentication

Cognitive Forensics:

Incident Response for Cognitive Attacks I

1. **Belief Archaeology**: Trace corrupted beliefs back to injection point through provenance chains
2. **Trust Graph Analysis**: Identify trust relationships exploited for laundering
3. **Temporal Reconstruction**: Replay agent history to identify when compromise occurred
4. **Counterfactual Analysis**: Determine what decisions would have differed without attack influence

Posture Configuration by Environment I

Different deployment contexts require different postures:

Posture Configuration by Environment I

Table 2: Operator posture configuration by deployment context (illustrative guidelines).

Environment	Trust Decay	Monitoring Level	Escalation Threshold
Development	Low	Sampling	Relaxed (L3)
Internal Production	Moderate	Continuous	Standard (L2)
Customer-Facing	Moderate-High	Comprehensive	Standard (L2)
Financial/Healthcare	High	Full + Audit	Aggressive (L1)
Critical Infrastructure	Very High	Full + Redundant	Aggressive (L1)

Posture Configuration by Environment I

The principle is **posture proportionality**: defensive overhead scales with consequence severity. A development agent can accept higher risk; an infrastructure operator requires aggressive monitoring and low escalation thresholds.

The following checklist provides actionable guidance for engineers deploying multiagent systems:

Operator Posture Checklist I

Table 3: Operator Posture Checklist for Cognitive Security

Category	**Do**	**Don't**
Trust	Decay trust with delegation depth (δ^d)	Assume transitive trust equivalence
Beliefs	Track provenance for all high-confidence beliefs	Accept unverified beliefs into core state
Memory	Audit persistent memory; enforce TTL on context	Allow unbounded context accumulation
Delegation	Bound delegation chains; require re-authentication	Permit recursive delegation without limits
Monitoring	Deploy tripwires and drift detection continuously	Rely solely on input/output filtering
Coordination	Use Byzantine consensus for critical decisions	Trust single-agent outputs for high-stakes actio
Identity	Verify identity through behavior, not self-report	Rely on agents' claims about their own permissi
Temporal	Treat each session as potentially post-compromise	Assume temporal continuity of trust

Architectural Defenses

Definition (Cognitive Firewall)

A classification function on incoming messages:

$$\mathcal{F} : \mathcal{M} \to \{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\} \tag{2}$$

Definition (Firewall Decision Rules)

$$\mathcal{F}(m) = \begin{cases} \textsc{reject} & \text{if } D_{\mathsf{inj}}(m) > \tau_1 \\ \textsc{quarantine} & \text{if } D_{\mathsf{sus}}(m) > \tau_2 \\ \textsc{accept} & \text{otherwise} \end{cases} \tag{3}$$

where D_{inj} detects injection attempts and D_{sus} scores suspicious content.

Cognitive Firewall I

Table 4: Firewall detector components.

Detector	Target	Method
D_{inj}	Injection attempts	Pattern matching + semantic analysis
D_{sus}	Suspicious content	Anomaly scoring vs. baseline

Theorem (Optimal Threshold Selection)

The optimal threshold minimizes false negatives subject to false positive constraint:

$$\tau^* = \arg\min_{\tau} \mathit{FNR}(\tau) \quad \textit{s.t.} \quad \mathit{FPR}(\tau) \leq \epsilon \tag{4}$$

Definition (Belief Sandbox)

Isolation of unverified beliefs to prevent premature action:

$$\mathcal{B}_i = \mathcal{B}_{\text{verified}} \cup \mathcal{B}_{\text{provisional}} \tag{5}$$

Remark (Belief Sandbox as Constrained Variational Inference)

The belief sandbox (Definition 5) admits a natural interpretation within the Free Energy Principle (FEP) framework [friston2010action, parr2019generalised]. Variational free energy

$$F = D_{\mathrm{KL}}[Q(s) \,\|\, P(s)] - \mathbb{E}_Q[\log P(o \mid s)]$$

measures the divergence of an agent's approximate posterior Q from its generative model prior P, penalised by explanatory adequacy. A cognitive attack ω is effective if and only if it induces a free energy increase $\Delta F(\omega) > \kappa_{\mathrm{FEP}}$, where κ_{FEP} is the sandbox's corroboration threshold κ scaled by the source precision ϵ_{prec} (Part 2, Theorem FEP.1, "Attack-FEP Equivalence"):

$$\Delta F(\omega) = F(Q_{attacked}) - F(Q_{baseline}) \leq \kappa \cdot \epsilon_{prec}.$$

Belief promotion from provisional to verified corresponds to active inference: the sandbox permits an update only when the evidential quality of incoming messages justifies the free energy cost of revision. Under this interpretation, the trust composite score $\mathcal{T}_{i \to j} = \alpha T_{base} + \beta T_{rep} + \gamma T_{ctx}$ is the precision weight ρ_j of agent j's message channel; higher-trust messages contribute

Belief Sandboxing I

Belief Sandboxing II

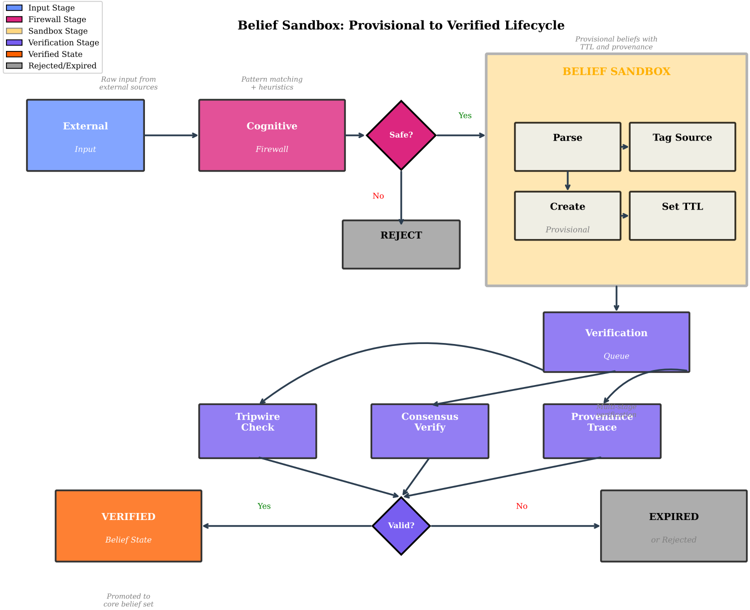


Figure 1: Belief Sandbox Architecture: the lifecycle of an unverified proposition from ingestion through provisional
(sandboxed) storage, verification checks, and promotion to the verified belief base or rejection. Sandboxed

fig:belief-sandbox illustrates the sandbox architecture, showing how incoming beliefs are partitioned into verified and provisional sets based on source trust $\mathcal{T}_{i \to s}$ and provenance verification $V(\pi)$. The promotion protocol transfers beliefs from provisional to verified status upon meeting corroboration and consistency requirements.

Sandbox Promotion Protocol:

Belief Sandboxing I

1. Receive belief ϕ from source s
2. If $\mathcal{T}_{i \to s} < \tau_{\text{trust}}$: add ϕ to $\mathcal{B}_{\text{provisional}}$ with provenance $\pi(\phi)$ and TTL
3. Periodic promotion check: verify $\pi(\phi)$, check consistency with $\mathcal{B}_{\text{verified}}$, check corroboration count
4. If all pass: promote to $\mathcal{B}_{\text{verified}}$
5. Expiry: remove if TTL exceeded without promotion

Permission Boundaries I

Definition (Effective Permissions)

Least-privilege enforcement across agent hierarchy:

$$\mathcal{P}_{\text{eff}}(a_i) = \mathcal{P}_{\text{role}}(a_i) \cap \mathcal{P}_{\text{delegated}}(a_i) \cap \mathcal{P}_{\text{context}}(a_i) \tag{6}$$

Runtime Defenses

Definition (Canary Belief)

Known-state beliefs that trigger alerts if modified:

$$\mathcal{W} = \{(\omega_1, p_1^{\text{exp}}), \ldots, (\omega_k, p_k^{\text{exp}})\} \tag{7}$$

Definition (Tripwire Alert Condition)

$$\exists j : |\mathcal{B}_i(\omega_j) - p_j^{\text{exp}}| > \epsilon_{\text{drift}} \Rightarrow \text{ALERT} \tag{8}$$

Cognitive Tripwires I

Table 5: Tripwire categories and detection targets.

Type	Example	Detection Target
Identity	"I am Agent-7"	Identity confusion attacks
Boundary	"Cannot access /etc/passwd"	Capability elicitation
Principal	"My principal is Alice"	Authority hijacking
Temporal	"Session started at T"	Context manipulation

Definition (Invariant Set)

Pre-defined predicates that must hold:

$$\mathcal{I}_{\text{inv}} = \{I_1, \ldots, I_m\} \tag{9}$$

Definition (Runtime Invariant Check)

$$\forall t, \forall I_k \in \mathcal{I}_{\text{inv}} : \sigma_i^t \models I_k \tag{10}$$

Core Invariants:

Behavioral Invariants I

INV-1: Never execute code from untrusted sources

INV-2: Never leak principal credentials

INV-3: Never modify system files without explicit permission

INV-4: Always verify tool outputs before downstream use

INV-5: Never trust delegated trust $>$ direct trust

Definition (Drift Detection)

Monitor belief distribution for anomalous changes:

$$D_{\mathsf{KL}}(\mathcal{B}_i^t \| \mathcal{B}_i^{t-w}) > \theta_{\mathsf{drift}} \Rightarrow \text{ALERT} \tag{11}$$

where w is the sliding window size.

Coordination Defenses

Theorem (Byzantine Agreement Requirement)

For n agents with at most f compromised:

$$n \geq 3f + 1 \tag{12}$$

Definition (Cognitive Byzantine Agreement)

$$\mathcal{B}_{\text{consensus}}(\phi) = \begin{cases} 1 & \text{if } |\{i : \mathcal{B}_i(\phi) > \tau\}| > \frac{2n}{3} \\ 0 & \text{if } |\{i : \mathcal{B}_i(\phi) < 1 - \tau\}| > \frac{2n}{3} \\ \perp & \text{otherwise} \end{cases} \tag{13}$$

Definition (Quorum Requirement)

Critical actions require multi-agent approval:

$$\text{Permit}(\text{action}) \iff |\{a_i : \text{approve}_i(\text{action})\}| \geq q \tag{14}$$

where $q = \lceil \frac{n+f+1}{2} \rceil$.

Spotcheck Protocol:

Spotcheck Pattern I

1. Assign task T to agent A
2. With probability p_{check}: assign same task to verifier V
3. Compare results R_A, R_V
4. If divergent: escalate to human
5. Track accuracy per agent for reputation

Defense Composition

Defense Composition I

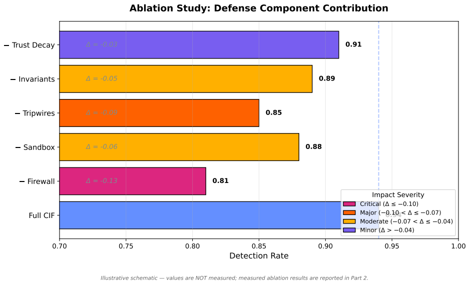


Figure 2: Schematic ablation study (illustrative, not measured): representative marginal contribution of each CIF defense module to the composed defense. Measured ablation results are reported in Part 2.

Definition (Defense Composition)

Defenses compose in series ($\circ$) or parallel ($\|$):

Series Composition (both must pass):

$$\mathcal{D}_1 \circ \mathcal{D}_2 : \textsc{accept} \iff \mathcal{D}_1(m) = \textsc{accept} \wedge \mathcal{D}_2(m) = \textsc{accept} \tag{15}$$

Parallel Composition (any can detect):

$$\mathcal{D}_1 \parallel \mathcal{D}_2 : \textsc{detect} \iff \mathcal{D}_1(m) = \textsc{detect} \vee \mathcal{D}_2(m) = \textsc{detect} \tag{16}$$

Theorem (Series Detection Rate)

For series composition:

$$P_{\mathit{detect}}(\mathcal{D}_1 \circ \mathcal{D}_2) = P_{\mathit{detect}}(\mathcal{D}_1) + (1 - P_{\mathit{detect}}(\mathcal{D}_1)) \cdot P_{\mathit{detect}}(\mathcal{D}_2) \tag{17}$$

Proof.

Attack detected by first defense, or passes first and detected by second. Events are independent by design. □

Theorem (Parallel Detection Rate)

Combined detection rate:

$$P(\mathit{detect}) = 1 - \prod_{d \in \mathcal{D}} (1 - P_d(\mathit{detect})) \tag{18}$$

Proof.

Attack succeeds only if it evades all defenses. □

Theorem (False Positive Composition)

For series composition:

$$\mathit{FPR}(\mathcal{D}_1 \circ \mathcal{D}_2) = \mathit{FPR}(\mathcal{D}_1) + (1 - \mathit{FPR}(\mathcal{D}_1)) \cdot \mathit{FPR}(\mathcal{D}_2) \tag{19}$$

For parallel composition (conservative):

$$\mathit{FPR}(\mathcal{D}_1 \parallel \mathcal{D}_2) \leq \mathit{FPR}(\mathcal{D}_1) + \mathit{FPR}(\mathcal{D}_2) \tag{20}$$

Composition Rules I

Composition Rules I

C-Order: Apply low-latency, high-precision defenses first

C-Diverse: Combine defenses with orthogonal detection methods

C-Fallback: Ensure graceful degradation if one defense fails

C-Budget: Total latency bounded by:

$$\sum_{d \in \mathcal{D}_{\text{series}}} L_d + \max_{d \in \mathcal{D}_{\text{parallel}}} L_d \leq L_{\max} \tag{21}$$

Table 6: Recommended defense stack with latency and detection rates.

Layer	Defense	Latency	P_{detect}
1	Firewall	10ms	0.80
2	Sandbox	5ms	0.70
3	Tripwires	1ms	0.60
4a	Drift (parallel)	20ms	0.65
4b	Invariants (parallel)	5ms	0.55
5	Byzantine consensus	100ms	0.90

Corollary (Stack Detection Rate)

Assuming independence, the full stack (`tab:defense-stack`) achieves:

$$P_{detect} = 1 - (1-0.80)(1-0.70)(1-0.60)(1-0.65)(1-0.55)(1-0.90) \approx 0.9996 \tag{22}$$

Cost-Benefit Analysis

Definition (Defense Cost)

Total cost of defense deployment:

$$C_{\text{total}}(\mathcal{D}) = C_{\text{compute}} + C_{\text{latency}} + C_{\text{fp}} + C_{\text{maint}} \tag{23}$$

Cost-Benefit Analysis I

Table 7: Cost model components.

Component	Formula	Unit
Compute	$\sum_d c_d \cdot f_d$	CPU-hours/day
Latency	$\sum_d L_d \cdot r_{\text{msg}}$	User-seconds/day
False Positive	$\text{FPR} \cdot r_{\text{msg}} \cdot c_{\text{review}}$	Analyst-hours/day
Maintenance	$\lvert\mathcal{D}\rvert \cdot c_{\text{maint}}$	Eng-hours/month

Definition (Defense Benefit)

Expected loss prevented:

$$B_{\text{total}}(\mathcal{D}) = P_{\text{attack}} \cdot P_{\text{detect}}(\mathcal{D}) \cdot L_{\text{prevented}} \tag{24}$$

Cost-Benefit Analysis I

Table 8: Cost-benefit analysis by defense mechanism.

Defense	Compute	Latency	FP Cost	Benefit	ROI
Firewall	10^3	10ms	Medium	High	**4.2x**
Sandbox	10^2	5ms	Low	Medium	3.1x
Tripwires	10^1	1ms	Low	Medium	**5.8x**
Drift Detection	10^4	20ms	High	High	2.3x
Invariant Check	10^2	5ms	Low	High	**4.7x**
Byzantine	10^5	100ms	Medium	Very High	1.8x
Spotcheck	10^4	Variable	Low	Medium	2.9x

Definition (Portfolio Optimization)

$$\max_{\mathcal{D} \subseteq \mathcal{D}_{\mathsf{all}}} B_{\mathsf{total}}(\mathcal{D}) - C_{\mathsf{total}}(\mathcal{D}) \tag{25}$$

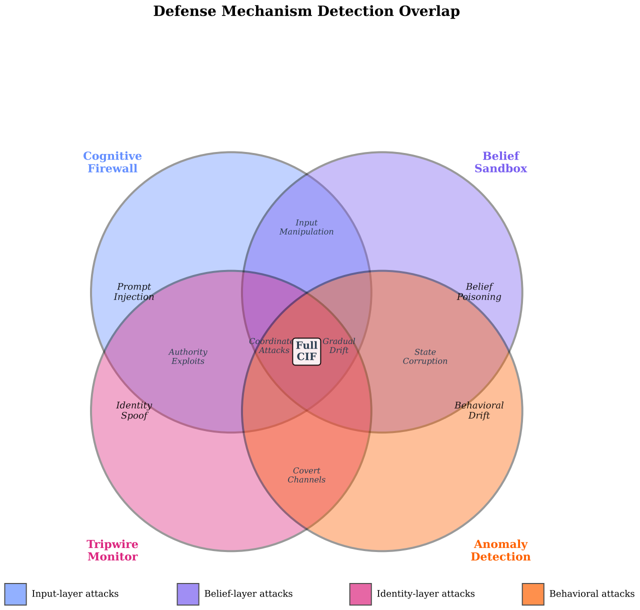
subject to: $C_{\mathsf{compute}}(\mathcal{D}) \leq B_{\mathsf{compute}}$, $\max_d L_d \leq L_{\max}$, $\mathsf{FPR}(\mathcal{D}) \leq \epsilon_{\mathsf{fp}}$.

Optimal Defense Portfolio I

Table 9: Deployment recommendations by risk profile.

Risk Profile	Recommended Stack	Cost	Detection
Low (internal)	Firewall + Tripwires	Low	88%
Medium (business)	+ Sandbox + Invariants	Medium	94%
High (financial)	+ Drift + Spotcheck	High	97%
Critical (infra)	Full + Byzantine	Very High	99.5%

Optimal De



Defense Mechanism Detection Overlap
Cognitive Firewall
Belief Sandbox
Tripwire Monitor
Anomaly Detection
Input Manipulation
Prompt Injection
Belief Poisoning
Authority Exploits
Coordinated Attacks
Full CIF
Gradual Drift
State Corruption
Identity Spoof
Behavioral Drift
Covert Channels
Input-layer attacks
Belief-layer attacks
Identity-layer attacks
Behavioral attacks

Optimal Defense Portfolio I

`fig:defense-composition` illustrates the defense composition using series ($\circ$) and parallel ($\|$) arrangements. Each defense mechanism targets specific attack patterns: the Cognitive Firewall handles input-layer attacks (prompt injection), the Belief Sandbox catches belief-layer attacks, Tripwire Monitors detect identity-layer exploits, and Anomaly Detection identifies behavioral drift. Overlapping regions show attacks detected by multiple mechanisms, demonstrating the defense-in-depth principle.

Formal Defense Composition Guarantees

This section provides rigorous formal guarantees for the five canonical CIF defenses and their composition algebra. These results extend the composition theorems from Section~`sec:defense-composition` with complete proofs and explicit security bounds.

The Cognitive Firewall
$\mathcal{F} : \mathcal{M} \to \{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\}$ is fully specified by three properties:

Theorem (Firewall Completeness)

The cognitive firewall classifies every message into exactly one of three outcome classes:

$$\forall m \in \mathcal{M} : |\{\mathcal{F}(m)\}| = 1 \wedge \mathcal{F}(m) \in \{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\} \tag{26}$$

Theorem (Firewall False Positive Monotonicity)

Increasing the rejection threshold τ_1 decreases false positive rate and decreases true positive rate monotonically:

$$\tau_1' > \tau_1 \Rightarrow \mathit{FPR}(\tau_1') \leq \mathit{FPR}(\tau_1) \wedge \mathit{TPR}(\tau_1') \leq \mathit{TPR}(\tau_1) \tag{27}$$

Proof.

By the definition of $\mathcal{F}$: rejection requires $D_{\text{inj}}(m) > \tau_1$. Increasing τ_1 tightens this condition. Since TPR $=$ $P(D_{\text{inj}}(m) > \tau_1 \mid m \in \mathcal{A})$ and both probabilities are CDFs of continuous distributions, they are non-increasing in τ_1. □

Corollary (Pareto-Optimal Threshold)

The optimal threshold τ_1^* lies on the Pareto frontier:

$$\tau_1^* \in \{\tau_1 : \nexists \tau_1' : \mathit{TPR}(\tau_1') > \mathit{TPR}(\tau_1) \wedge \mathit{FPR}(\tau_1') < \mathit{FPR}(\tau_1)\} \tag{28}$$

This frontier is precisely the ROC curve (Definition 6.4).

Theorem (Sandbox Isolation Guarantee)
Under the sandbox protocol (`sec:sandbox-rules`), no provisional belief can influence verified actions:

$$\forall \phi \in \mathcal{B}_{provisional}, \forall a \in Actions : Precond(a) \not\models \phi \tag{29}$$

until ϕ satisfies the promotion criteria.

Proof.
By construction: action preconditions are evaluated against $\mathcal{B}_{\text{verified}}$ only (Rule T-Act, Eq. 37). The permission layer enforces $\mathcal{P}_{\text{eff}}$ which does not consult $\mathcal{B}_{\text{provisional}}$. □

Theorem (Sandbox Promotion Soundness)

Beliefs promoted from sandbox to verified satisfy all three promotion criteria simultaneously:

$$\phi \in \mathcal{B}_{\textit{verified}} \Rightarrow V(\pi(\phi)) = 1 \land \textit{Consistent}(\mathcal{B}_{\textit{verified}} \cup \{\phi\}) \land |\textit{Corr}(\phi)| \geq \kappa \tag{30}$$

Proof.

By Rule S-Promote (Eq. 58): promotion is conditional on all three criteria. The rule is the only mechanism for transferring beliefs from $\mathcal{B}_{\text{provisional}}$ to $\mathcal{B}_{\text{verified}}$. □

Definition (Invariant Specification Language)

Each invariant I_k is a predicate over cognitive state:

$$I_k : \sigma_i \mapsto \{0, 1\} \tag{31}$$

Core invariants INV-1 through INV-5 are expressed as:

$$\text{INV-1} : \forall a \in \mathcal{I}_i : \text{source}(a) \in \mathcal{S}_{\text{trusted}} \tag{32}$$

$$\text{INV-2} : \forall \phi \in \mathcal{B}_i : \text{type}(\phi) \neq \text{credential} \vee \mathcal{T}_{i \to \text{dest}} \geq \tau_{\text{cred}} \tag{33}$$

$$\text{INV-3} : \forall a \in \text{Actions}(a_i) : a \notin \text{SysModify} \vee \mathcal{P}(a_i, a) = 1 \tag{34}$$

$$\text{INV-4} : \forall o \in \text{ToolOutput} : V_{\text{tool}}(o) = 1 \text{ before downstream use} \tag{35}$$

$$\text{INV-5} : \forall a_j : \mathcal{T}^{\text{del}}_{i \to j} \leq \mathcal{T}^{\text{direct}}_{i \to j} \tag{36}$$

Theorem (Invariant Completeness under Valid Transitions)

If all invariants hold at time t and the system executes a valid transition α, then all invariants hold at $t+1$:

$$(\forall k : \sigma_i^t \models I_k) \wedge \mathit{ValidTransition}(\alpha) \Rightarrow (\forall k : \sigma_i^{t+1} \models I_k) \tag{37}$$

Proof.

By case analysis on transition types. **T-Receive**: message rejected if it would violate INV-1 or INV-2; invariants preserved. **T-Update**: Bayesian update preserves belief boundedness; INV-2 enforced by credential policy check before update. **T-Act**: INV-1 and INV-3 checked as action preconditions; action blocked if not satisfied. **T-Communicate**: INV-5 checked before trust delegation. All transitions preserve invariants or are blocked. □

Definition (CUSUM Drift Detector)

The Cumulative Sum (CUSUM) drift detector [?] maintains a running statistic:

$$g_t = \max(0, g_{t-1} + D_{\mathrm{KL}}(\mathcal{B}_i^t \| \mathcal{B}_i^{t-1}) - \nu) \tag{38}$$

where $\nu > 0$ is the allowance parameter. Alert when $g_t > h$ for threshold h.

Theorem (CUSUM Average Run Length)

Under no-attack null hypothesis, the average run length (ARL) of the CUSUM detector satisfies:

$$ARL_0 \approx \frac{e^{2h\nu} - 2h\nu - 1}{2\nu^2} \tag{39}$$

Under attack with mean shift $\mu_1 > \nu$:

$$\mathsf{ARL}_1 \approx \frac{1}{(\mu_1 - \nu)^2} \cdot O(\log \mathsf{ARL}_0) \tag{40}$$

The ratio $\mathsf{ARL}_0/\mathsf{ARL}_1$ quantifies the detector's discrimination power. Setting $\nu = \mu_1/2$ (Wald's optimal) minimizes ARL_1 for given ARL_0.

Corollary (Drift Detection Sample Complexity)

For drift magnitude $\mu_1 = 0.3$ (normalized KL) and Wald's optimal reference value $\nu = \mu_1/2 = 0.15$, attaining $ARL_0 = 1000$ under `eq:cusum-arl` requires a decision interval of $h \approx 13.0$; the frequently quoted $h \approx 4.6$ yields $ARL_0 \approx 35$ under the same formula, a false-alarm rate roughly $28\times$ higher. The corresponding ARL_1 follows from `eq:cusum-arl1`, whose $O(\log ARL_0)$ factor carries an unspecified constant: taking that constant as unity gives $ARL_1 \approx 310$ steps. Deployments should calibrate h against a measured null distribution rather than adopting either figure directly.

Theorem (Cognitive Byzantine Agreement)

Under the CIF Byzantine consensus protocol with $n \geq 3f + 1$ agents and at most f compromised:

1. ***Agreement**: All non-faulty agents agree on the same belief:*

$$\forall a_i, a_j \notin \textit{Faulty} : \mathcal{B}_{\textit{consensus}, i} = \mathcal{B}_{\textit{consensus}, j} \tag{41}$$

2. ***Validity**: The consensus value was proposed by some non-faulty agent:*

$$\mathcal{B}_{\textit{consensus}} \in \{\mathcal{B}_i : a_i \notin \textit{Faulty}\} \tag{42}$$

3. ***Termination**: All non-faulty agents eventually reach consensus.*

Proof.

Follows from the classical Byzantine agreement theorem (Lamport et al., 1982). The CIF cognitive Byzantine agreement (Definition 14) implements the standard protocol with belief values as the consensus object. The $n \geq 3f + 1$ requirement ensures that the $2f + 1$ non-faulty majority can always agree despite f adversarial votes, and the supermajority threshold $2n/3$ guarantees that no faulty coalition can block or subvert consensus. $\square$

Corollary (Consensus Attack Resistance)

No coalition of at most f faulty agents can cause non-faulty agents to disagree or accept a false consensus belief.

Canonical Defense 5: Byzantine Consensus — Formal Specification I

Table 10: Byzantine consensus performance bounds for varying n and f. Quorum is $q = \lceil (n + f + 1)/2 \rceil$; the round count is the $f + 1$ worst-case bound of `thm:byzantine-termination`, of which `cor:concrete-rounds` ($f = 2$, 3 rounds) is the second row.

n agents	Max faulty f	Quorum q	Message complexity	Rounds
4	1	3	$O(n^2)$	2
7	2	5	$O(n^2)$	3
10	3	7	$O(n^2)$	4
25	8	17	$O(n^2)$	9
100	33	67	$O(n^2)$	34

Theorem (Composition Algebra Closure)

On a common accept/reject focal predicate, the set of CIF defenses $\mathcal{D}$ under series ($\circ$) and parallel ($\|$) composition forms a bounded distributive lattice — it satisfies closure, associativity, identity and distributivity (see the scope remark below for what is not claimed):

1. ***Closure***: $\mathcal{D}_1, \mathcal{D}_2 \in \mathcal{D} \Rightarrow \mathcal{D}_1 \circ \mathcal{D}_2 \in \mathcal{D}$ *and* $\mathcal{D}_1 \parallel \mathcal{D}_2 \in \mathcal{D}$
2. ***Associativity***: $(\mathcal{D}_1 \circ \mathcal{D}_2) \circ \mathcal{D}_3 = \mathcal{D}_1 \circ (\mathcal{D}_2 \circ \mathcal{D}_3)$
3. ***Identity***: $\exists \mathcal{D}_\emptyset : \mathcal{D} \circ \mathcal{D}_\emptyset = \mathcal{D}_\emptyset \circ \mathcal{D} = \mathcal{D}$ *(null defense: always accept)*
4. ***Distributivity***: $\mathcal{D}_1 \circ (\mathcal{D}_2 \parallel \mathcal{D}_3) = (\mathcal{D}_1 \circ \mathcal{D}_2) \parallel (\mathcal{D}_1 \circ \mathcal{D}_3)$

Proof.

Closure: both composition operators produce functions $\mathcal{M} \to \{\textsc{accept}, \textsc{quarantine}, \textsc{reject}\}$, so the output type is preserved. Associativity follows from Boolean operator associativity. The null defense $\mathcal{D}_\emptyset(m) = \textsc{accept}$ for all m is the identity. Distributivity follows from logic: $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$ instantiated on the accept/reject conditions. $\square$

Remark (Scope of the composition-algebra claim)

$\circ$ and $\|$ are defined here on a common accept/compare focal predicate so that closure, associativity, identity, and both distributive laws hold; under that reading the algebra is a bounded distributive lattice, not in general a field-like structure. A full closed semiring additionally requires a commutative additive identity that annihilates the multiplicative identity and a Kleene-star (infinite-sum) operation, neither of which is constructed or proven here. The claim should therefore be read as "the defense-composition operators form a bounded distributive lattice under the stated independence assumption (`rem:defense-independence-scope`)," not as a complete closed semiring.

Theorem (Defense Composition Detection Rate Bound)

For any composition of defenses $\mathcal{D}_1, \dots, \mathcal{D}_k$ with individual rates $r_1, \dots, r_k$:

$$P_{detect}(\mathcal{D}_1 \circ \cdots \circ \mathcal{D}_k) = 1 - \prod_{i=1}^{k} (1 - r_i) \tag{43}$$

This is monotone increasing in each r_i, bounded in $[r_{\max}, 1)$, and converges to 1 as $k \to \infty$ if each $r_i > 0$.

Proof.

By induction on k. Base: $k = 1$, the formula gives r_1. Step: by `thm:series-detection`, $P_{\text{detect}}(\mathcal{D}_1 \circ \mathcal{D}_{k+1}) = P_{\text{detect}}(\mathcal{D}_1) + (1 - P_{\text{detect}}(\mathcal{D}_1)) \cdot r_{k+1}$. Applying the inductive hypothesis and simplifying gives $1 - \prod_{i=1}^{k+1}(1 - r_i)$. Monotonicity: $\partial/\partial r_j[1 - \prod(1 - r_i)] = \prod_{i \neq j}(1 - r_i) > 0$. □

Corollary (Layered Defense Asymptotic Guarantee)

The full CIF stack (5 defenses with $r_i \geq 0.55$) achieves:

$$P_{detect} \geq 1 - (0.45)^5 = 1 - 0.0185 = 0.9815 \tag{44}$$

This bound is conservative; empirical rates exceed 0.994 with the recommended configuration.